

BOL'SHERECHYE, Russian FederationWMO: **285930**Lat: **56.0941N**Lon: **74.6103E**Elev: **254**StdP: **14.56**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.6	-26.8	-37.9	0.6	-32.3	-32.1	0.9	-26.5	19.3	18.2	17.1	13.6	1.8	210	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	84.9	66.8	81.7	65.5	78.8	64.1	69.6	80.2	67.9	77.9	66.2	75.2	7.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	96.5	74.2	64.2	90.8	73.0	62.5	85.5	71.2	33.9	80.3	32.5	77.9	31.1	75.2	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	16.2	14.1	DB	-37.0	89.2	8.1	3.5	-42.8	91.7	-47.5	93.8	-52.1	95.8	-57.9	98.3	(4)
(5)			WB	-37.1	72.3	8.0	1.7	-42.9	73.5	-47.6	74.5	-52.1	75.4	-57.9	76.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.3	-1.3	2.6	19.9	38.0	52.5	62.8	64.8	60.3	49.8	36.9	19.2	4.7		
	DBStd	25.82	14.75	13.59	10.41	10.77	9.68	7.95	6.64	7.09	8.03	8.90	13.30	14.78		
	HDD50	7177	1591	1328	935	381	87	7	0	5	100	412	925	1406		
	HDD65	11392	2056	1748	1400	810	399	133	87	182	461	870	1375	1871		
	CDD50	1460	0	0	0	21	166	389	460	324	93	7	0	0		
	CDD65	200	0	0	0	0	13	66	82	35	4	0	0	0		
	CDH74	1992	0	0	0	8	209	669	729	333	44	0	0	0		
	CDH80	468	0	0	0	1	45	169	175	74	4	0	0	0		
	WSAvg	6.3	5.6	6.3	7.0	7.2	7.0	5.8	5.0	5.3	5.9	6.7	7.1	6.6		
	PrecAvg	15.9	0.7	0.5	0.6	0.9	1.5	2.2	3.0	2.0	1.3	1.1	1.2	0.9		
	PrecMax	25.3	1.6	0.9	1.5	4.5	3.2	5.0	8.7	4.6	3.3	2.0	2.3	1.6		
(15) Precipitation	PrecMin	11.0	0.0	0.0	0.0	0.0	0.5	0.4	0.4	0.4	0.4	0.3	0.2	0.3		
	PrecStd	3.3	0.4	0.2	0.4	0.8	0.8	1.0	1.7	1.1	0.7	0.5	0.6	0.4		
	0.4%	DB	31.9	34.7	43.4	73.8	85.1	88.1	89.0	86.4	79.2	64.6	44.1	34.6		
	2%	DB	28.4	30.7	39.3	65.3	79.4	84.5	84.5	81.2	73.1	59.0	40.4	32.0		
	5%	DB	25.0	27.5	36.7	59.9	75.2	81.3	81.2	77.5	67.6	54.1	37.1	28.3		
(23) Mean Coincident Wet Bulb Temperatures	10%	DB	20.6	22.7	34.3	54.9	70.4	77.7	78.2	73.8	63.7	49.9	34.1	24.3		
	MCWB	19.6	21.3	31.6	45.6	55.9	63.8	65.8	62.6	54.0	43.8	32.5	23.2	23.2		
	0.4%	WB	30.6	32.1	37.8	56.8	64.2	71.7	72.3	70.0	63.5	52.8	41.7	33.8		
	2%	WB	27.2	29.1	35.4	52.1	61.7	68.9	70.2	68.0	59.7	50.1	38.6	30.5		
	5%	WB	23.8	26.0	33.6	49.3	59.2	66.8	68.5	65.9	57.2	47.3	35.3	27.0		
(31) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	WB	19.6	21.4	31.8	46.1	56.9	64.8	66.9	63.9	54.7	44.5	32.8	23.1		
	MCDB	20.5	22.7	34.3	54.5	68.3	75.1	75.7	71.9	62.2	49.1	34.0	24.2	24.2		
	MDBR	13.7	15.7	15.9	17.1	21.2	19.1	17.8	18.1	16.7	12.6	10.7	13.1	13.1		
	5% DB	MCDBR	13.6	12.8	14.1	25.0	28.0	25.0	23.0	24.1	23.5	20.4	10.1	12.6		
	5% WB	MCWBR	13.1	12.1	10.9	14.4	12.8	10.8	9.6	11.1	11.9	12.1	9.3	12.3		
(38) Clear-Sky Solar Irradiance	taub	0.280	0.313	0.372	0.410	0.401	0.410	0.428	0.423	0.368	0.353	0.316	0.283	0.283		
	taud	2.036	2.014	1.923	2.076	2.232	2.283	2.230	2.217	2.332	2.220	2.076	1.978	1.978		
	Ebn at Noon	186	227	240	253	266	265	256	248	247	213	167	146	146		
	Edh at Noon	20	31	44	44	41	39	40	38	30	26	20	17	17		
	RadAvg	268	605	1094	1446	1684	1859	1707	1347	926	458	218	169	169		
	RadStd	22	35	70	121	162	147	115	127	93	51	38	21	21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page